EC571: Digital VLSI Circuit Design

Homework 6 — Fall 2026

Total: 100 points

Due: November 15, 2026 at 11:59 pm ET

General Instructions

Use this worksheet to complete the homework. Please print this worksheet, write your solutions on the worksheet, take a picture of the worksheet, and submit it through Gradescope. If you prefer, you can download the Homework 6 worksheet on your tablet, complete the worksheet on your tablet, and submit the completed worksheet through Gradescope.

Show all reasoning. A final numerical answer without intermediate steps will receive limited credit.

There are five questions, each worth 20 points. Answer all questions. State the assumptions and circuit operating conditions that justify your calculations. Draw and label circuit diagrams wherever requested. Use the transistor and timing models covered in Lectures 12–15; select the relevant equations yourself.

You may use course notes, lecture slides, and a calculator. You may also be required to do some research to answer some of the questions. AI tools may be used for concept review but not to generate final solutions. Include a brief AI-use disclosure with your submission.

Write your responses in the blank space provided under each part; attach additional pages through Gradescope if necessary.

Question 1: SRAM Read Stability and Read Delay [20 points]

Consider a conventional six-transistor (6T) SRAM cell storing $Q = 0$ and $\overline{Q} = 1$. The bitline attached to Q is denoted BL, and the complementary bitline attached to $\overline{Q}$ is denoted $\overline{BL}$. Both bitlines are initially precharged to $V_{DD} = 1.0\,\text{V}$. The wordline is initially LOW and then asserted HIGH for a read.

For this problem, use a simplified ON-resistance model during the initial part of the read. On the Q side, the access transistor has an ON resistance of $12\,\text{k}\Omega$. Three alternative pull-down transistor sizes are under consideration; their corresponding ON resistances are shown below. Assume that the low storage node rapidly reaches the voltage established by the two ON resistances while BL is still close to its precharged voltage. Ignore leakage and body effect.

Candidate	A	B	C
Pull-down width (normalized)	1	2	3
Pull-down ON resistance	$12\,\text{k}\Omega$	$6\,\text{k}\Omega$	$4\,\text{k}\Omega$

As a simplified read-stability requirement, the low storage node must remain *strictly below* $0.30\,\text{V}$ throughout the initial read disturbance. This is a specified design criterion for this problem, not a direct measurement of butterfly-curve static noise margin.

For a separate first-order timing estimate, model the path discharging BL as the series combination of the access and pull-down resistances. The total capacitance of BL is $180\,\text{fF}$, initially charged to $1.0\,\text{V}$. The sense amplifier can make a reliable decision after BL falls by $100\,\text{mV}$. Neglect the storage-node capacitance and wordline delay for this timing estimate.

(a) **Read mechanism.** [4 points]

Draw and label the 6T SRAM cell, its two access transistors, and the differential bitlines. Identify the current path when the wordline is asserted and explain which bitline begins to discharge.

(b) **Read-disturb evaluation.** **[6 points]**

Estimate the initial disturbed voltage at the internal node Q for candidates A, B, and C. Identify every candidate that satisfies the read-stability requirement. Explain how the cell ratio affects the result.

(c) **Bitline sensing time.** **[6 points]**

For each candidate, estimate the time required for BL to fall by 100 mV after the wordline is asserted, using the stated timing model. Identify the fastest candidate that also meets the stability requirement.

(d) **Design tradeoff.** **[4 points]**

Explain how strengthening the pull-down transistor affects read stability, storage-node capacitance, and cell area. Discuss how a larger cell could indirectly affect array interconnect parasitics. Would a faster sense amplifier, by itself, eliminate an initial read-upset problem? Explain.

Question 2: SRAM Write Ability and Transistor-Strength Tradeoffs [20 points]

Consider a 6T SRAM cell initially storing $Q = 1$ and $\overline{Q} = 0$. The goal is to write a 0 into Q. The write drivers force $BL = 0$ and $\overline{BL} = 1.0\,\text{V}$, after which the wordline is asserted. The supply is $V_{DD} = 1.0\,\text{V}$.

Before the cross-coupled inverters switch state, assume the pull-up PMOS connected to Q behaves as an ON resistance R_{PU}, and the access NMOS connecting Q to BL behaves as an ON resistance of $10\,\text{k}\Omega$. The opposite inverter switches when Q is pulled *strictly below* $0.40\,\text{V}$. This is a simplified switching criterion; neglect the storage-node capacitance during the initial voltage-divider analysis.

Candidate	A	B	C
Pull-up ON resistance R_{PU}	$8\,\text{k}\Omega$	$16\,\text{k}\Omega$	$24\,\text{k}\Omega$

For the read-stability check in part (c), the pull-down NMOS on the side storing 0 has an ON resistance of $4\,\text{k}\Omega$, both bitlines are precharged to $1.0\,\text{V}$, and the same $0.40\,\text{V}$ internal-node limit is used.

(a) **Write contention.** **[5 points]**

Sketch the relevant transistors and voltages during the beginning of the write operation. Explain which devices oppose each other and why a write can fail even though BL is driven strongly to ground.

(b) **Pull-up selection.** **[6 points]**

Estimate the pre-switching value of Q for each pull-up candidate. Identify which candidates meet the specified write-switching criterion. For the candidates that work, calculate the approximate direct-path current from V_{DD} to the low bitline while this initial contention persists.

(c) **Access-transistor upsizing.** **[5 points]**

Suppose the access NMOS is upsized so that its ON resistance decreases from $10\,\mathrm{k}\Omega$ to $5\,\mathrm{k}\Omega$, with the pull-down and pull-up devices unchanged. For candidate B, re-evaluate the initial write voltage at Q and the disturbed low-node voltage during a read. Does the larger access transistor help or hurt each operation? Does it satisfy the specified read-voltage limit?

(d) **Final cell recommendation.** **[4 points]**

Choose a pull-up candidate and an access-transistor option for a cell that must meet both the read and write requirements. Justify the choice in terms of write ability, read disturb, contention current, cell area, and input loading. Explain one limitation of the simplified resistance model.

Question 3: SRAM Array Organization, Decoder, and Read Access Time [20 points]

A designer must implement a 65,536-bit, single-bit-addressable SRAM macro using 6T cells. The macro has a row decoder that selects one wordline and a column multiplexer that selects one differential bitline pair for the output. Consider three organizations:

Organization	Number of rows	Number of columns
A	128	512
B	256	256
C	512	128

Use the following first-order model, with $V_{DD} = 1.0\,\mathrm{V}$:

- The selected wordline has a fixed capacitance of 40 fF plus 0.25 fF per cell in its row. The equivalent row-driver resistance is 2.4 kΩ.
- Each bitline has a fixed capacitance of 25 fF plus 0.40 fF per cell connected along its column. The selected cell discharges the bitline through an effective constant resistance of 14 kΩ.
- The selected bitline is initially at 1.0 V, and the sense amplifier responds when it reaches 0.90 V.
- Row-decoder delay is 80 ps; the combined column-multiplexer and sense-amplifier delay is 85 ps, independent of the array organization.
- Treat the wordline and bitline as lumped RC networks, neglect wire resistance, and assume the bitline discharge begins after the wordline reaches its 50% point. All listed delays are serial for this model.

(a) **Addressing and selection.** [4 points]

For each organization, determine the number of row-address and column-address bits. Explain the function of the row decoder and column multiplexer. Confirm that all three organizations store the required number of bits.

(b) **Interconnect loading.** **[5 points]**

Compute the selected wordline capacitance and bitline capacitance for each organization. Which organization has the largest wordline load? Which has the largest bitline load? Explain physically.

(c) **Read-access comparison.** **[7 points]**

Estimate the total read-access time for each organization, including row decoding, wordline charging to the 50% level, bitline development of the required sense voltage, and column selection/sensing. Rank the organizations from fastest to slowest.

(d) **Sensitivity to parasitics.** **[4 points]**

Suppose routing increases the bitline capacitance contribution per attached cell from 0.40 fF to 0.65 fF without changing any other parameters. Re-evaluate the fastest organization. Explain why the preferred array aspect ratio can change with the interconnect assumptions.

Question 4: Choosing a 16-bit Adder Architecture [20 points]

A 16-bit datapath must add two unsigned operands and produce the sum and final carry. The designer is considering three architectures from Lecture 14: a ripple-carry adder, a carry-bypass (carry-skip) adder, and a linear carry-select adder. For the latter two, equal-sized groups are used and the allowable group sizes are 2, 4, and 8 bits.

Use the lecture's first-order critical-path conventions and these delay parameters:

Timing component	Delay
Operand generate/propagate setup	30 ps
One carry-propagation step	24 ps
Final sum generation	35 ps
One block-bypass operation	45 ps
One carry-select multiplexer operation	40 ps

Assume the setup work can be performed in parallel across the operand bits, ignore wire delay and other parasitics, and use the same setup delay for all three architectures. Compare the path to the most-significant sum bit using the timing paths described in Lecture 14.

(a) **Ripple-carry baseline.** [4 points]

Identify the longest carry-dependent path and estimate the 16-bit ripple-carry adder delay. Explain why its delay grows with operand width.

(b) **Carry-bypass design space.** **[6 points]**

Estimate the critical-path delay of carry-bypass adders with group sizes of 2, 4, and 8 bits. Show the carry/bypass steps you count and identify the best group size among these options.

(c) **Carry-select design space.** **[6 points]**

Repeat the timing comparison for a linear carry-select architecture with group sizes of 2, 4, and 8 bits. Identify the best group size and compare the fastest carry-select implementation with the fastest carry-bypass implementation.

(d) **Timing versus implementation cost.** **[4 points]**

Recommend one architecture for (i) the minimum delay and (ii) a design with a 370 ps delay budget but strong area/energy constraints. Justify the answers, accounting for duplicated carry computations, multiplexers, regularity, and switching activity. State which tradeoffs cannot be quantified from the given timing data alone.

Question 5: Multiplier Critical Paths and Partial-Product Reduction [20 points]

Consider unsigned multiplication. Use the conventional array, carry-save multiplier, and Wallace-tree organizations introduced in Lecture 15. In parts (b)–(c), both operands are 8 bits wide, and the relevant timing parameters are:

Timing component	Delay
Partial-product AND operation	20 ps
One carry-propagation step in multiplier array	30 ps
One sum-propagation step in multiplier array	45 ps
Final vector-merging adder in carry-save multiplier	230 ps

For the conventional array and carry-save multiplier, follow the *critical-path counting conventions shown in Lecture 15*, including the contribution of partial-product generation along the modeled path. Ignore wiring and loading differences in the numerical estimates.

(a) **Partial-product organization.** [4 points]

For a 4×4 unsigned multiplication, draw the partial-product rows aligned by bit weight. Show the number of partial-product bits in each product column (from weight 0 through weight 6). Explain how adders reduce these bits to the eight product bits.

(b) **Conventional array delay.** **[6 points]**

Using the conventional array architecture and its critical-path model from lecture, estimate the critical-path delay of an 8×8 multiplier. Explicitly identify the numbers of modeled carry, sum, and AND-delay contributions and explain where they occur.

(c) **Carry-save alternative.** **[5 points]**

Estimate the delay of an 8×8 carry-save multiplier with the specified final vector-merging adder. Compare its delay with the conventional array and determine the percentage delay reduction. How would the result change if the vector-merging adder were redesigned to have a 140 ps delay?

(d) **Wallace-tree design reasoning.** **[5 points]**

Using your 4×4 partial-product columns from part (a), illustrate one legal first-stage Wallace-tree compression step. Mark at least one full adder, show the destination weight of each sum and carry output, and explain why the tree can reduce the number of serial reduction levels. Discuss one disadvantage of a tree relative to a regular array multiplier.